Supplementary Material

This document provides supplementary material for the manuscript "Sleep and the Daytime Course of Anhedonia in Adolescent Depression: An Ecological Momentary Assessment Study" by Hannah E. Marshall, Linda Nduka, Tahjanee Givens, Caroline Miller, Mollie Davis, Chana Engel, Kenneth Towbin, Daniel Pine, and Katharina Kircanski.

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Supplement S1

Deviations From Preregistration

The current manuscript contains slight deviations from our preregistration (https://osf.io/r685d/), detailed below:

1. In the preregistration, we planned to operationalize sleep quality using reverse-scored ratings of sleep disturbances. Upon further consideration, we determined that this operationalization required an inferential step that was not sufficiently supported by the measure itself. Therefore, we instead used the original, non-reverse-scored ratings and focused our analyses and interpretation directly on sleep disturbances rather than making inferences about sleep quality.
2. In our manuscript, we rephrased our hypotheses to reflect sleep disturbances as opposed to sleep quality. The measures of interest and hypothesized directions of effects remained consistent with our preregistration.
3. Our preregistered analysis plan involved re-running our analyses controlling for tiredness. For succinctness and parsimony, we have omitted these results from the manuscript but have included them in Tables S19–22. For information on how tiredness was measured, see Appendix S2.

Supplement S2

Tiredness Measurement

Tiredness was assessed using a single EMA item: *I felt tired*. Adolescents rated the item on sliding visual analogue scale, with underlying values ranging from 0–100, where 0 indicated "not at all" and 100 indicated "extremely." Participants reported tiredness since the previous prompt at every EMA prompt (i.e., four times per day).

Table S1

Descriptive Statistics and Intraclass Correlation Coefficients

	Major Depressive Disorder								Healthy Volunteer							
Measure	*M*	SD_W	SD_B	Min_W	Min_B	Max_W	Max_B	ICC	*M*	SD_W	SD_B	Min_W	Min_B	Max_W	Max_B	ICC
Sleep disturbances	48.38	23.48	17.65	10.68	1.00	87.66	78.19	.32	18.20	16.14	13.90	3.29	0.00	59.01	55.69	.38
Sleep duration	38.54	15.71	12.32	10.01	6.59	70.62	62.71	.33	44.35	14.56	8.83	18.63	27.93	71.05	65.75	.24
Anticipatory anhedonia	50.53	21.50	14.13	11.79	12.69	94.75	78.75	.27	25.25	15.19	15.25	4.00	2.04	79.37	52.46	.47
Consummatory anhedonia	50.12	21.59	11.23	9.21	12.82	93.70	71.34	.19	27.54	14.50	16.34	4.46	2.09	78.28	65.71	.53

Note. M is the mean of person-level means. W denotes within-person metrics. B denotes between-person metrics. SD_W is the mean of the person-level standard deviations. SD_B is the standard deviation of the person-level means. Min_W is the mean person-level minimum. Min_B is the minimum of the person-level means. Max_W is the mean person-level maximum. Max_B is the maximum of the person-level means.

Table S2

Within-Person and Between-Person Posterior Correlations Between Variables

	Major Depressive Disorder						Healthy Volunteer					
Measure	r_W	95% CrI	BF_{10}	r_B	95% CrI	BF_{10}	r_W	95% CrI	BF_{10}	r_B	95% CrI	BF_{10}
Sleep disturbances–sleep duration	-0.29	-0.34, -0.23	6.32×10^{20}	-0.15	-0.46, 0.18	0.32	-0.27	-0.34, -0.19	1.30×10^{17}	-0.02	-0.44, 0.39	0.28
Anticipatory anhedonia–consummatory anhedonia	0.49	0.47, 0.52	5.77×10^{14}	0.90	0.82, 0.95	2.43×10^{15}	0.31	0.27, 0.35	8.92×10^{14}	0.89	0.76, 0.95	2.13×10^{15}

Note. W denotes within-person metrics. B denotes between-person metrics. r_W is the mean of person-level posterior median correlation estimates. r_B is the posterior median correlation estimate of person-level mean ratings.

Table S3

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	3.740	2.219	-0.557, 8.180	0.049	**0.951****
Sleep disturbances	0.101	0.099	-0.091, 0.296	0.158	0.842
Time	-0.170	0.210	-0.591, 0.241	0.792	0.208
Group (HV)	-3.024	3.987	-10.915, 4.790	0.778	0.222
Sleep disturbances × Time	-0.004	0.010	-0.024, 0.015	0.662	0.338
Sleep disturbances × Group (HV)	-0.371	0.226	-0.802, 0.068	**0.952****	0.048
Time × Group (HV)	0.090	0.398	-0.676, 0.893	0.405	0.595
Sleep disturbances ×Time × Group (HV)	0.034	0.023	-0.010, 0.078	0.067	**0.933***

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S7. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S4

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	6.158	2.544	1.600, 11.872	0.004	**0.996****
Sleep disturbances	-0.104	0.102	-0.289, 0.101	0.840	0.160
Time	-0.370	0.250	-0.954, 0.062	**0.952****	0.048
Group (HV)	-3.519	4.955	-17.267, 4.866	0.777	0.223
Sleep disturbances × Time	0.018	0.010	-0.003, 0.036	0.042	**0.958****
Sleep Disturbances × Group (HV)	-0.207	0.230	-0.660, 0.252	0.816	0.184
Time × Group (HV)	0.232	0.508	-0.627, 1.677	0.325	0.675
Sleep Disturbances × Time × Group (HV)	0.017	0.023	-0.030, 0.064	0.224	0.776

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S8. Consummatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. ***** Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ****** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S5

Anticipatory Anhedonia as a Function of the Sleep Duration ×Time of Day × Group Interaction, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	3.496	2.175	-0.819, 7.733	0.057	**0.943***
Sleep duration	-0.194	0.145	-0.480, 0.090	**0.910***	0.090
Time	-0.151	0.206	-0.553, 0.258	0.775	0.225
Group (HV)	-1.760	3.982	-9.583, 5.948	0.666	0.334
Sleep duration × Time	0.008	0.013	-0.017, 0.033	0.257	0.743
Sleep duration × Group (HV)	0.414	0.292	-0.165, 0.983	0.078	**0.922***
Time × Group (HV)	-0.020	0.401	-0.798, 0.772	0.528	0.472
Sleep duration × Time × Group (HV)	-0.029	0.028	-0.084, 0.027	0.851	0.149

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S9. Anticipatory anhedonia and fsleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S6

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	4.928	2.358	0.433, 9.705	0.016	**0.984****
Sleep duration	-0.312	0.139	-0.580, -0.036	**0.987****	0.013
Time	-0.259	0.224	-0.718, 0.166	0.879	0.121
Group (HV)	-2.442	4.093	-10.540, 5.568	0.726	0.274
Sleep duration × Time	0.015	0.012	-0.010, 0.038	0.119	0.881
Sleep duration × Group (HV)	0.380	0.281	-0.164, 0.945	0.086	**0.914***
Time × Group (HV)	0.153	0.411	-0.634, 0.976	0.357	0.643
Sleep duration × Time × Group (HV)	-0.027	0.027	-0.081, 0.027	0.832	0.168

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S10. Consummatory anhedonia and sleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S7

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	0.741	3.205	-5.479, 7.189	0.412	0.589
Sleep disturbances	-0.272	0.209	-0.684, 0.136	**0.902***	0.098
Time	-0.083	0.332	-0.746, 0.566	0.594	0.406
Group (MDD)	3.062	3.838	-4.445, 10.730	0.216	0.784
Sleep disturbances × Time	0.030	0.021	-0.012, 0.072	0.077	**0.923***
Sleep disturbances × Group (MDD)	0.370	0.231	-0.087, 0.829	0.056	**0.944***
Time × Group (MDD)	-0.091	0.388	-0.853, 0.677	0.590	0.410
Sleep disturbances × Time × Group (MDD)	-0.034	0.023	-0.080, 0.013	**0.926***	0.074

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S8

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	3.024	3.415	-3.759, 9.614	1.181	0.819
Sleep disturbances	-0.292	0.206	-0.690, 0.112	**0.922***	0.078
Time	-0.176	0.353	-0.871, 0.526	0.698	0.302
Group (MDD)	2.923	4.136	-5.191, 11.279	0.235	0.765
Sleep disturbances × Time	0.033	0.021	-0.008, 0.074	0.059	**0.941***
Sleep disturbances × Group (MDD)	0.189	0.230	-0.265, 0.647	0.203	0.797
Time × Group (MDD)	-0.173	0.418	-1.013, 0.638	0.658	0.342
Sleep disturbances ×Time × Group (MDD)	-0.016	0.024	-0.061, 0.030	0.749	0.251

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Consummatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S9

Anticipatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.697	3.408	-4.811, 8.287	0.306	0.694
Sleep duration	0.225	0.249	-0.262, 0.720	0.182	0.818
Time	-0.167	0.354	-0.845, 0.512	0.686	0.314
Group (MDD)	1.849	4.120	-6.200, 10.365	0.333	0.667
Sleep duration × Time	-0.021	0.024	-0.070, 0.027	0.812	0.188
Sleep duration × Group (MDD)	-0.420	0.290	-0.988, 0.152	**0.928***	0.072
Time × Group (MDD)	0.010	0.417	-0.874, 0.812	0.490	0.510
Sleep duration × Time × Group (MDD)	0.029	0.028	-0.026, 0.083	0.142	0.858

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Anticipatory anhedonia and sleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S10

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	2.478	3.468	-4.218, 9.183	0.231	0.769
Sleep duration	0.072	0.242	-0.394, 0.539	0.389	0.611
Time	-0.102	0.356	-0.797, 0.595	0.617	0.383
Group (MDD)	2.465	4.212	-5.782, 11.093	0.277	0.723
Sleep duration × Time	-0.012	0.024	-0.060, 0.035	0.692	0.308
Sleep duration × Group (MDD)	-0.383	0.282	-0.940, 0.162	**0.916***	0.084
Time × Group (MDD)	-0.156	0.421	-1.011, 0.667	0.649	0.351
Sleep duration × Time × Group (MDD)	0.027	0.028	-0.027, 0.081	0.168	0.833

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Consummatory anhedonia and sleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S11

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Sex and Age, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.668	6.296	-10.769, 13.981	0.399	0.601
Sleep disturbances	0.099	0.097	-0.089, 0.291	0.154	0.847
Time	-0.162	0.203	-0.547, 0.238	0.789	0.211
Group (HV)	-2.275	3.942	-9.895, 5.445	0.718	0.282
Sex (M)	-1.691	1.174	-3.982, 0.643	**0.923***	0.077
Age	0.012	0.032	-0.049, 0.074	0.355	0.645
Sleep disturbances × Time	-0.004	0.010	-0.023, 0.015	0.645	0.354
Sleep disturbances × Group (HV)	-0.355	0.225	-0.801, 0.075	**0.945***	0.056
Time × Group (HV)	0.071	0.393	-0.712, 0.834	0.429	0.571
Sleep disturbances × Time × Group (HV)	0.032	0.023	-0.011, 0.078	0.077	**0.923***

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S15. Females were the reference sex, and males (M) were the comparison sex. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S12

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Sex and Age, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	10.087	7.031	-3.833, 24.166	0.075	**0.925***
Sleep disturbances	-0.105	0.105	-0.299, 0.105	0.844	0.156
Time	-0.309	0.228	-0.766, 0.125	**0.918***	0.082
Group (HV)	-1.721	4.296	-10.263, 6.419	0.651	0.349
Sex (M)	-2.457	1.356	-5.101, 0.219	**0.961****	0.039
Age	-0.021	0.035	-0.091, 0.049	0.732	0.268
Sleep disturbances × Time	0.017	0.010	-0.003, 0.037	0.048	**0.952****
Sleep disturbances × Group (HV)	-0.160	0.240	-0.637, 0.292	0.742	0.258
Time × Group (HV)	0.115	0.435	-0.722, 0.959	0.401	0.599
Sleep disturbances × Time × Group (HV)	0.013	0.024	-0.033, 0.062	0.306	0.694

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S16. Females were the reference sex, and males (M) were the comparison sex. Consummatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S13

Anticipatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Sex and Age, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.339	6.387	-11.799, 13.568	0.416	0.584
Sleep duration	-0.199	0.121	-0.433, 0.040	**0.952****	0.048
Time	-0.110	0.205	-0.521, 0.274	0.697	0.303
Group (HV)	-1.016	3.890	-8.852, 6.483	0.606	0.394
Sex (M)	-1.805	1.214	-4.177, 0.597	**0.930***	0.070
Age	0.011	0.032	-0.050, 0.073	0.372	0.628
Sleep duration × Time	0.009	0.010	-0.011, 0.028	0.197	0.803
Sleep duration × Group (HV)	0.416	0.267	-0.107, 0.939	0.059	0.941
Time × Group (HV)	-0.027	0.387	-0.772, 0.735	0.532	0.468
Sleep duration × Time × Group (HV)	-0.030	0.026	-0.081, 0.021	0.881	0.119

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S17. Females were the reference sex, and males (M) were the comparison sex. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S14

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Sex and Age, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	9.465	7.332	-5.036, 23.197	0.096	**0.904***
Sleep duration	-0.360	0.126	-0.611, -0.116	**0.997****	0.003
Time	-0.197	0.230	-0.653, 0.242	0.797	0.203
Group (HV)	-0.905	4.245	-9.138, 7.249	0.580	0.420
Sex (M)	-2.566	1.363	-5.323, 0.082	**0.971****	0.029
Age	-0.024	0.037	-0.097, 0.052	0.748	0.252
Sleep duration × Time	0.019	0.010	-0.001, 0.040	0.035	**0.965****
Sleep duration × Group (HV)	0.437	0.275	-0.105, 0.953	0.054	**0.947***
Time × Group (HV)	0.068	0.421	-0.732, 0.912	0.446	0.554
Sleep duration × Time × Group (HV)	-0.031	0.027	-0.081, 0.024	0.886	0.114

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S18. Females were the reference sex, and males (M) were the comparison sex. Consummatory anhedonia and sleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S15

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Sex and Age, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	-0.608	6.523	-13.232, 12.183	0.535	0.465
Sleep disturbances	-0.251	0.206	-0.654, 0.154	0.888	0.112
Time	-0.098	0.328	-0.744, 0.542	0.622	0.378
Group (MDD)	2.210	3.841	-5.389, 9.797	0.277	0.723
Sex (M)	-1.715	1.193	-4.022, 0.614	**0.925***	0.075
Age	0.013	0.032	-0.048, 0.073	0.343	0.657
Sleep disturbances × Time	0.028	0.021	-0.013, 0.069	0.089	**0.911***
Sleep disturbances × Group (MDD)	0.347	0.229	-0.106, 0.806	0.064	**0.935***
Time × Group (MDD)	-0.065	0.384	-0.819, 0.687	0.570	0.430
Sleep disturbances × Time × Group (MDD)	-0.032	0.023	-0.078, 0.014	**0.913***	0.086

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Females were the reference sex, and males (M) were the comparison sex. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S16

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Sex and Age, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	7.736	7.493	-6.784, 22.544	0.154	0.847
Sleep disturbances	-0.285	0.205	-0.674, 0.120	**0.917***	0.083
Time	-0.164	0.355	-0.853, 0.553	0.678	0.322
Group (MDD)	2.138	4.197	-5.821, 10.633	0.304	0.696
Sex (M)	-2.426	1.402	-5.169, 0.404	**0.958****	0.043
Age	-0.019	0.036	-0.090, 0.050	0.696	0.304
Sleep disturbances × Time	0.032	0.021	-0.010, 0.073	0.064	**0.936***
Sleep disturbances × Group (MDD)	0.182	0.228	-0.269, 0.618	0.214	0.786
Time × Group (MDD)	-0.162	0.420	-1.018, 0.660	0.652	0.348
Sleep disturbances × Time × Group (MDD)	-0.015	0.024	-0.061, 0.032	0.736	0.264

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Females were the reference sex, and males (M) were the comparison sex. Consummatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S17

Anticipatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Sex and Age, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	0.319	6.605	-12.828, 13.309	0.478	0.521
Sleep duration	0.213	0.232	-0.247, 0.666	0.176	0.824
Time	-0.146	0.336	-0.811, 0.506	0.667	0.333
Group (MDD)	0.973	3.925	-6.733, 8.623	0.405	0.595
Sex (M)	-1.804	1.205	-4.157, 0.588	**0.933***	0.067
Age	0.011	0.031	-0.049, 0.073	0.362	0.638
Sleep duration × Time	-0.021	0.023	-0.067, 0.025	0.817	0.183
Sleep duration × Group (MDD)	-0.410	0.263	-0.925, 0.099	**0.943***	0.057
Time × Group (MDD)	0.036	0.391	-0.717, 0.802	0.463	0.537
Sleep duration × Time × Group (MDD)	0.029	0.025	-0.020, 0.079	0.125	0.875

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Females were the reference sex, and males (M) were the comparison sex. Anticipatory anhedonia and sleep disturbances were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S18

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Sex and Age, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	8.473	7.746	-6.989, 23.309	0.136	0.864
Sleep duration	0.073	0.230	-0.376, 0.516	0.379	0.621
Time	-0.117	0.377	-0.851, 0.621	0.625	0.375
Group (MDD)	1.174	4.409	-7.210, 9.939	0.404	0.596
Sex (M)	-2.564	1.442	-5.404, 0.201	**0.966****	0.034
Age	-0.024	0.037	-0.099, 0.049	0.739	0.262
Sleep duration × Time	-0.012	0.023	-0.057, 0.034	0.692	0.308
Sleep duration × Group (MDD)	-0.425	0.258	-0.926, 0.079	**0.950****	0.050
Time × Group (MDD)	-0.096	0.444	-0.979, 0.754	0.581	0.420
Sleep duration × Time × Group (MDD)	0.030	0.025	-0.019, 0.079	0.116	0.884

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Females were the reference sex, and males (M) were the comparison sex. Consummatory anhedonia and sleep duration were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S19

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Tiredness, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.274	2.214	-2.974, 5.574	0.286	0.715
Sleep disturbances	0.108	0.099	-0.081, 0.300	0.133	0.867
Time	0.003	0.207	-0.405, 0.397	0.490	0.510
Group (HV)	-3.401	3.863	-10.865, 4.140	0.811	0.189
Tiredness	0.166	0.024	0.120, 0.213	0.000	**1.000****
Sleep disturbances × Time	-0.006	0.010	-0.025, 0.013	0.720	0.280
Sleep disturbances × Group (HV)	-0.434	0.220	-0.873, -0.003	**0.975****	0.025
Time × Group (HV)	0.144	0.388	-0.608, 0.889	0.353	0.647
Sleep disturbances × Time × Group (HV)	0.039	0.022	-0.005, 0.084	0.037	**0.963****

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S23. Anticipatory anhedonia, sleep disturbances, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S20

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Tiredness, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	4.420	2.320	-0.270, 8.835	0.035	**0.965****
Sleep disturbances	-0.102	0.103	-0.312, 0.096	0.845	0.155
Time	-0.238	0.215	-0.654, 0.188	0.863	0.137
Group (HV)	-2.815	4.287	-10.942, 4.944	0.739	0.261
Tiredness	0.107	0.023	0.061, 0.150	0.000	**1.000****
Sleep disturbances × Time	0.017	0.010	-0.002, 0.038	0.045	**0.955****
Sleep disturbances × Group (HV)	-0.236	0.228	-0.681, 0.212	0.849	0.151
Time × Group (HV)	0.171	0.432	-0.589, 1.010	0.349	0.651
Sleep disturbances × Time × Group (HV)	0.020	0.023	-0.025, 0.065	0.195	0.805

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S24. Consummatory anhedonia, sleep disturbances, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S21

Anticipatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Tiredness, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.264	2.182	-3.038, 5.556	0.287	0.713
Sleep duration	-0.184	0.147	-0.472, 0.094	0.899	0.101
Time	-0.002	0.204	-0.405, 0.394	0.505	0.495
Group (HV)	-1.917	3.918	-9.530, 5.746	0.689	0.311
Tiredness	0.169	0.024	0.122, 0.216	0.000	**1.000****
Sleep duration × Time	0.010	0.013	-0.014, 0.036	0.212	0.787
Sleep duration × Group (HV)	0.481	0.288	-0.069, 1.058	0.047	**0.953****
Time × Group (HV)	0.010	0.398	-0.777, 0.792	0.486	0.514
Sleep duration × Time × Group (HV)	-0.035	0.028	-0.088, 0.019	0.898	0.102

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S25. Anticipatory anhedonia, sleep duration, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S22

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Tiredness, with Adolescents with MDD as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	3.510	2.320	-1.013, 8.177	0.066	**0.934***
Sleep duration	-0.304	0.141	-0.575, -0.025	**0.983****	0.017
Time	-0.161	0.217	-0.587, 0.262	0.776	0.224
Group (HV)	-2.548	4.167	-10.752, 5.484	0.727	0.273
Tiredness	0.106	0.025	0.058, 0.153	0.000	**1.000****
Sleep duration × Time	0.016	0.013	-0.009, 0.040	0.100	**0.900***
Sleep duration × Group (HV)	0.419	0.277	-0.129, 0.956	0.068	**0.931***
Time × Group (HV)	0.173	0.417	-0.613, 0.986	0.344	0.656
Sleep duration × Time × Group (HV)	-0.030	0.027	-0.083, 0.024	0.862	0.138

Note. Adolescents with major depressive disorder were the reference group, and healthy volunteers (HV) were the comparison group. For results from the corresponding model with HV as the reference group, see Table S26. Consummatory anhedonia, sleep duration, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. ***** Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ****** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S23

Anticipatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Tiredness, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	-2.124	3.165	-8.345, 4.102	0.752	0.248
Sleep disturbances	-0.329	0.203	-0.723, 0.067	**0.949***	0.052
Time	0.146	0.329	-0.506, 0.801	0.327	0.673
Group (MDD)	3.435	3.762	-3.847, 10.851	0.183	0.817
Tiredness	0.166	0.023	0.120, 0.213	0.000	**1.000****
Sleep disturbances × Time	0.033	0.021	-0.007, 0.074	0.052	**0.948***
Sleep disturbances × Group (MDD)	0.435	0.225	-0.016, 0.877	0.029	**0.971****
Time × Group (MDD)	-0.146	0.379	-0.892, 0.585	0.648	0.352
Sleep disturbances × Time × Group (MDD)	-0.039	0.023	-0.084, 0.007	**0.952****	0.048

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Anticipatory anhedonia, sleep disturbances, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S24

Consummatory Anhedonia as a Function of the Sleep Disturbances × Time of Day × Group Interaction, Controlling for Tiredness, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.057	3.437	-5.662, 7.706	0.377	0.623
Sleep disturbances	-0.331	0.206	-0.725, 0.074	**0.947***	0.053
Time	-0.016	0.352	-0.712, 0.675	0.519	0.481
Group (MDD)	3.328	4.189	-4.801, 11.531	0.213	0.787
Tiredness	0.106	0.024	0.059, 0.153	0.000	**1.000****
Sleep disturbances × Time	0.035	0.021	-0.006, 0.075	0.048	**0.952****
Sleep disturbances × Group (MDD)	0.233	0.229	-0.212, 0.684	0.158	0.843
Time × Group (MDD)	-0.225	0.417	-1.046, 0.582	0.711	0.289
Sleep disturbances × Time × Group (MDD)	-0.019	0.023	-0.064, 0.026	0.791	0.209

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Consummatory anhedonia, sleep disturbances, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S25

Anticipatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Tiredness, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	-0.633	3.248	-6.963, 5.634	0.571	0.429
Sleep duration	0.299	0.250	-0.180, 0.798	0.113	0.887
Time	0.006	0.335	-0.640, 0.670	0.497	0.503
Group (MDD)	1.866	3.877	-5.682, 9.616	0.318	0.682
Tiredness	0.169	0.024	0.121, 0.217	0.000	**1.000****
Sleep duration × Time	-0.024	0.025	-0.073, 0.023	0.838	0.162
Sleep duration × Group (MDD)	-0.486	0.289	-1.064, 0.081	**0.956****	0.044
Time × Group (MDD)	-0.006	0.391	0.778, 0.771	0.510	0.490
Sleep duration × Time × Group (MDD)	0.035	0.028	-0.019, 0.091	0.101	0.899

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Anticipatory anhedonia, sleep duration, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.

Table S26

Consummatory Anhedonia as a Function of the Sleep Duration × Time of Day × Group Interaction, Controlling for Tiredness, with Healthy Volunteers as the Reference Group

Predictor	β	SE	95% CrI	P(β < 0)	P(β > 0)
Intercept	1.074	3.515	-5.894, 7.652	0.381	0.619
Sleep duration	0.115	0.241	-0.362, 0.583	0.317	0.683
Time	-0.003	0.359	-0.681, 0.703	0.503	0.497
Group (MDD)	2.398	4.270	-5.977, 10.921	0.287	0.713
Tiredness	0.107	0.024	0.061, 0.154	0.000	**1.000****
Sleep duration × Time	-0.014	0.024	-0.061, 0.034	0.718	0.282
Sleep duration × Group (MDD)	-0.417	0.280	-0.958, 0.141	**0.930***	0.070
Time × Group (MDD)	-0.158	0.429	-1.017, 0.663	0.637	0.362
Sleep duration × Time × Group (MDD)	0.030	0.027	-0.024, 0.083	0.139	0.861

Note. Healthy volunteers were the reference group, and adolescents with major depressive disorder (MDD) were the comparison group. Consummatory anhedonia, sleep duration, and tiredness were person-mean centered. Time of day was operationalized as hours past midnight. * Moderate evidence; P(β < 0) > 0.900 | P(β > 0) > 0.900. ** Strong evidence; P(β < 0) > 0.950 | P(β > 0) > 0.950.